Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6.	(a)	(i)		Substitution into: $v^2 = u^2 + 2ax$ i.e. $v^2 = 225 - 2 \times 3.5 \times 20$ (1) $v^2 = 85$ (1) $v = 9.22$ [m/s] (1) **Award 2 marks** for an answer of 19.1 [m/s] **Award 1 mark** for an answer of 365 [m/s]	1	1 1		3	3	
		(ii)		5 625 – 2 125 = 3 500 (1) Substitution: 3 500 **ecf** = $F \times 20$ (1) N.B. the ecf only applies to an incorrect subtraction $F = \frac{3\,500}{20} = 175$ [N] (1) **Award 2 marks**: $F = 281.25$ [N] **or** $F = 106.25$ [N] **or** $F = 193.75$ [N]	1	1 1		3	3	
	(b)			Increase the distance (1) over which the energy is transferred / [same] work is done / work done = force × distance (1) so the force decreases (1) **Or** Increase the time (1) so there is decreased deceleration / to reduce the momentum / force = $\frac{\text{change in momentum}}{\text{time}}$ (1) so the force decreases (1) Don't accept impact	3			3		
				Question 6 total	**5**	**4**	**0**	**9**	**6**	**0**

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